

10X Genomics scRNAseq of Monocularly Inactivated *Macaca mulatta* (rhemac10)

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Questions for Via Scientific

1. Thoughts on current analysis pipeline? Are there any concerns or things we should do differently?
2. Are there other validations you imagine a reviewer would want to see?
3. Could VS help us create a pipeline to “integrate” our datasets with others so we can impose those cell-type labels or validate our cell-type classification :
 - E.g. our data and only other published Macaque V1 dataset:
 - Wei et al. 2022: <https://www.nature.com/articles/s41467-022-34590-1>
DOI: <https://doi.org/10.1038/s41467-022-34590-1>
 - *Note: Wei et al used *macaque fascicularis* (not *mulatta*) so we just used homologous genes
4. Are there other ways in which VS thinks we could predict cell activation state?

Background-1

- Neuronal activity-regulated gene expression is essential for healthy brain development, function, memory, and plasticity.
- Mutations in several genes that are part of this program cause severe neurodevelopmental disorders (transcription factors, chromatin regulators, ion channels).
- GWAS have repeatedly associated loci involved activity-dependent gene program with autism, schizophrenia, depression, bipolar, intelligence, educational attainment, etc..
- Our previous studies found significant enrichment of:
 - 1- autism-associated genes in activity-regulated genes of stem cell-derived neurons
 - 2- autism risk in activity-regulated promoters of stem cell-derived neurons (LDSC)

Background-2

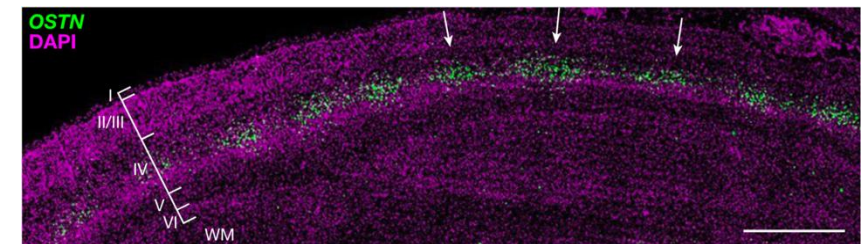
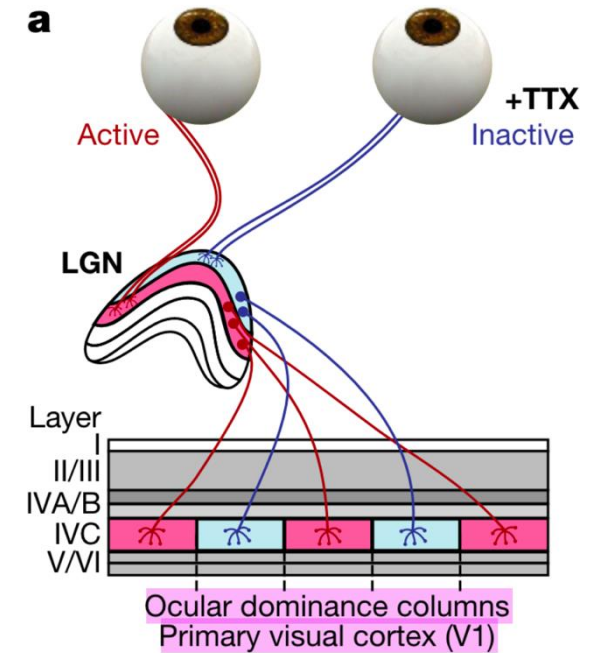
- Neuronal activity-regulated gene expression programs have been extensively studied in rodent *in vivo*.
- Neuronal activity-regulated (nAD) gene expression programs in the rodent brain are cell-type specific, including in non-neuronal cells.
- From cell culture studies, our lab has found dozens of nAD genes that are specific to primate (including humans) and are likely part of primate-specific pathways for neuronal plasticity.
- There has previously not been a way to study nAD gene programs in primate or human *in vivo*.
- This will be the first *in vivo* study directly examining nAD gene programs in primate.

Goal

Discover novel primate neuronal activity-regulated gene expression at cell-type resolution *in vivo*.

Overview

- Doing single nuclear RNA-seq on the primary visual cortex (V1) of Rhesus macaque (*Macaca mulatta*) monkeys after Monocular Inactivation (MI) (TTX injection blinds one eye), which produces active and inactive ocular dominance columns (ODC) in primary visual cortex (V1). 5 individual brains banked.
- So far 20 libraries prepared with 10X Chromium Next GEM Single Cell 3' Kit v3.1, ~ 10k nuclei per library
- 7 libraries sequenced with Illumina NextSeq 2000 (14 paired-end FASTQ files) → 56k nuclei pass QC



Ataman et al. 2016

Experimental Plan

1. Obtain approx. 200k high-quality nuclei from all libraries sequenced

2. Finalize clustering and ID cell types

→ Integrate dataset with Wei et al. (2022). Identification of visual cortex cell types and species differences using single-cell RNA sequencing. *Nature Communications*, 13, Article 6902. <https://doi.org/10.1038/s41467-022-34590-1>

3. Identify activated vs. inactivated transcriptional profile for each neuronal cell type

→ Predict “active” cells based on expression of small number of known activity-regulated genes in:

- All Excitatory neurons
- Layer IV excitatory neurons
- All Inhibitory Neurons
- VIP neurons (doi: 10.1038/nature17187)
- Any subtypes for which we can reliably call “active” vs “inactive” nuclei

→ Validate using RNA FISH in tissue sections

4. Identify novel neuronal activity-dependent gene expression patterns by comparing Macaque activity-dependent transcriptomes with those from Mouse V1 scRNAseq (DOI: [10.1038/s41593-017-0029-5](https://doi.org/10.1038/s41593-017-0029-5))

Initial Input data

- Species: Rhesus macaque (*Macaca mulatta*)
- Conditions
 - MI1- monocular inactivation animal 1
 - MI2- monocular inactivation animal 2
- Replicates
 - 7 technical replicates sequenced and analyzing to start (data presented here)
 - 5 reps from MI1
 - 2 reps from MI2
 - 15 libraries sequenced yet unmapped
 - 3 libraries yet unsequenced

```
○ 1-MI1_230808_L1_ds.016079dff48748459f0a59413df9957c
  ▪ 1-MI1_230808_S3_L001_R1_001.fastq.gz
  ▪ 1-MI1_230808_S3_L001_R2_001.fastq.gz
○ 2-MI1_230808_L1_ds.5fea222a7e2c4f758af8ac4c096722d2
  ▪ 2-MI1_230808_S4_L001_R1_001.fastq.gz
  ▪ 2-MI1_230808_S4_L001_R2_001.fastq.gz
○ 3-C_MI2_230608_L1_ds.27acae0b835f4cb9bd014cb36b5c2a39
  ▪ 3-C_MI2_230608_S1_L001_R1_001.fastq.gz
  ▪ 3-C_MI2_230608_S1_L001_R2_001.fastq.gz
○ 4-D_MI2_230608_L1_ds.35850839ff214617925e9c5d1e694bbe
  ▪ 4-D_MI2_230608_S2_L001_R1_001.fastq.gz
  ▪ 4-D_MI2_230608_S2_L001_R2_001.fastq.gz
○ 4-MI1_230808_L1_ds.c5b578e5de6041c98332dbfb7424c4e6
  ▪ 4-MI1_230808_S5_L001_R1_001.fastq.gz
  ▪ 4-MI1_230808_S5_L001_R2_001.fastq.gz
○ 5-MI1_230808_L1_ds.72e56be26126499c8bebfbd669a1e78
  ▪ 5-MI1_230808_S6_L001_R1_001.fastq.gz
  ▪ 5-MI1_230808_S6_L001_R2_001.fastq.gz
○ 7-MI1_230808_L1_ds.3f0f802bee2547ae86cb94d85cd11570
  ▪ 7-MI1_230808_S7_L001_R1_001.fastq.gz
  ▪ 7-MI1_230808_S7_L001_R2_001.fastq.gz
```


10x Requirements for Genome Reference

- 10 Tech Support:

"If the species is available from the [Ensembl database](#), we recommend using the files from there. The GTF files from Ensembl contain optional tags that make filtering easy. If your species of interest is not available from Ensembl, GTF and FASTA files from other sources can also work. Note that a GTF file is required, while a GFF file is not supported. (See [GFF/GTF File Format - Definition and supported options](#))"

 - Ensembl GTF from rhemac10 missing activity-dependent genes we are interested in (*see next slides for annotation comparisons*)
 - We are unable to use rhemac10 NCBI Annotation with Cell Ranger (wrong format)
 - Software has been tested and validated only using Ensembl references

Rhemac10 Annotation Comparisons

Genome annotation	File name	Number of unique genes
NCBI	GCF_003339765.1_Mmul_10_genomic.gtf	40,307
Ensembl	Macaca_mulatta.Mmul_10.105.gtf	17,542
UCSC	rheMac10.ncbiRefSeq.gtf	33,690
Liftoff NCBI RefSeq	rheMac10_liftoff_GRCh38.p13_RefSeq.gtf	37,252
Liftoff GENCODEv47 Comprehensive	rheMac10-hg38-gencode.v47.comp.gtf	69,513
Liftoff GENCODEv47 Basic	rheMac10-hg38-gencode.v47.basic.gtf	69,554

*Results in later slides are from using rheMac10_liftoff_GRCh38.p13_RefSeq.gtf before GENCODE liftoff's were made

Used rhemac10 liftoff annotation & UCSC assembly

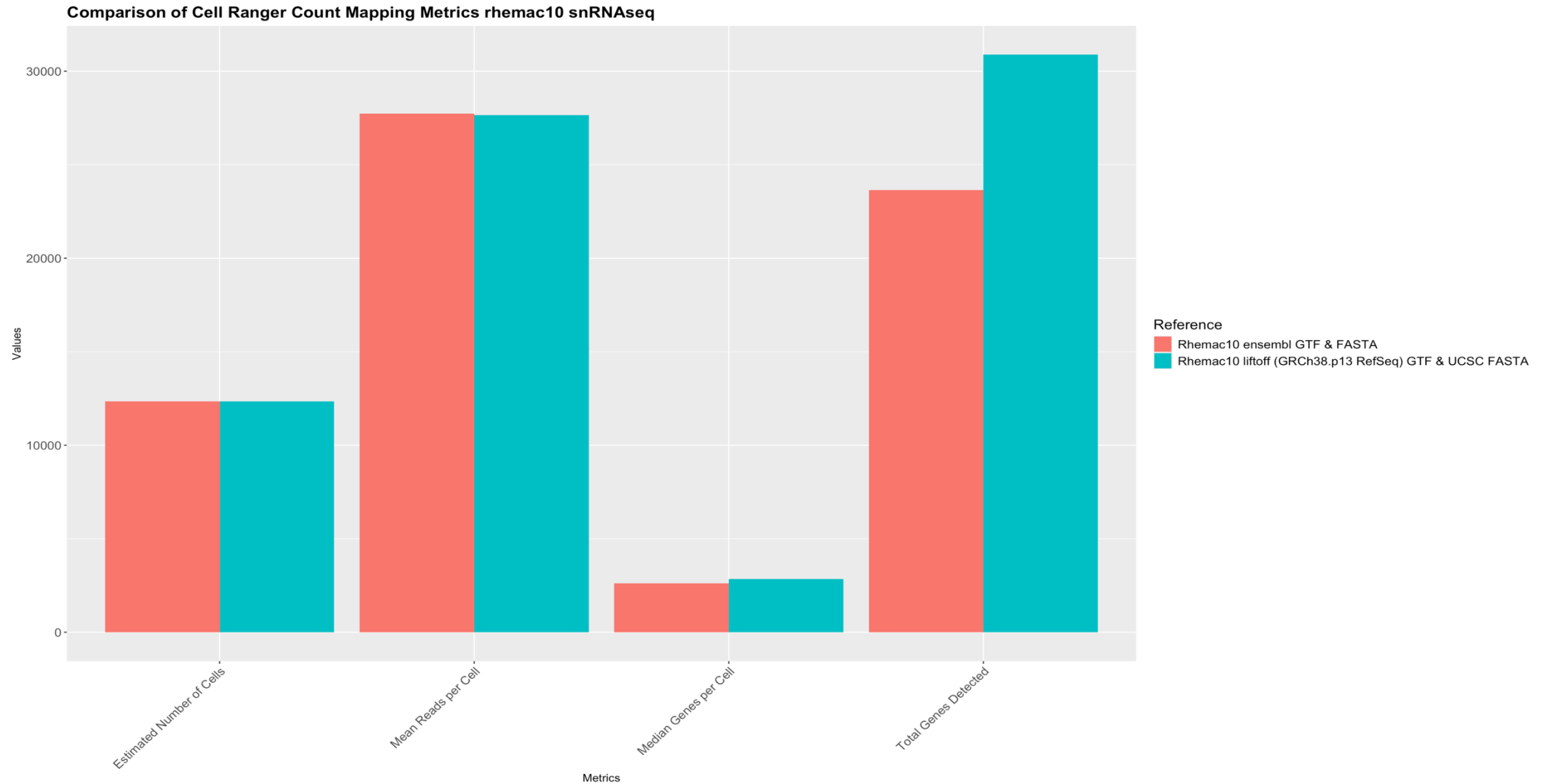
- Use [liftoff](#) software to make a more complete rhemac10 genome annotation GTF file using the most recent human genome annotation
 - Liftoff takes in 2 FASTA files as input (rhemac10 FASTA & GRCh38.p13 FASTA) + a reference genome annotation (GRCh38.p13 GTF) and outputs a target genome annotation GTF of the target genome (rhemac10 GTF). Software aligns gene sequences to be “lifted over” onto a new genome annotation
 - We used the rhemac10 annotation gtf file (rheMac10_liftoff_GRCh38.p13_RefSeq.gtf) that was created using liftoff by Phan et al. (2024). Single nuclei transcriptomics in human and non-human primate striatum in opioid use disorder. *Nature Communications*, 15, Article 878. <https://doi.org/10.1038/s41467-024-45165-7>
 - Deposited to [Carnegie Mellon University's Kilthub repository resource](#)
- Used rhemac10 FASTA from UCSC Genome Browser
<https://hgdownload.soe.ucsc.edu/goldenPath/rheMac10/bigZips/rheMac10.2bit>

snRNAseq analysis pipeline

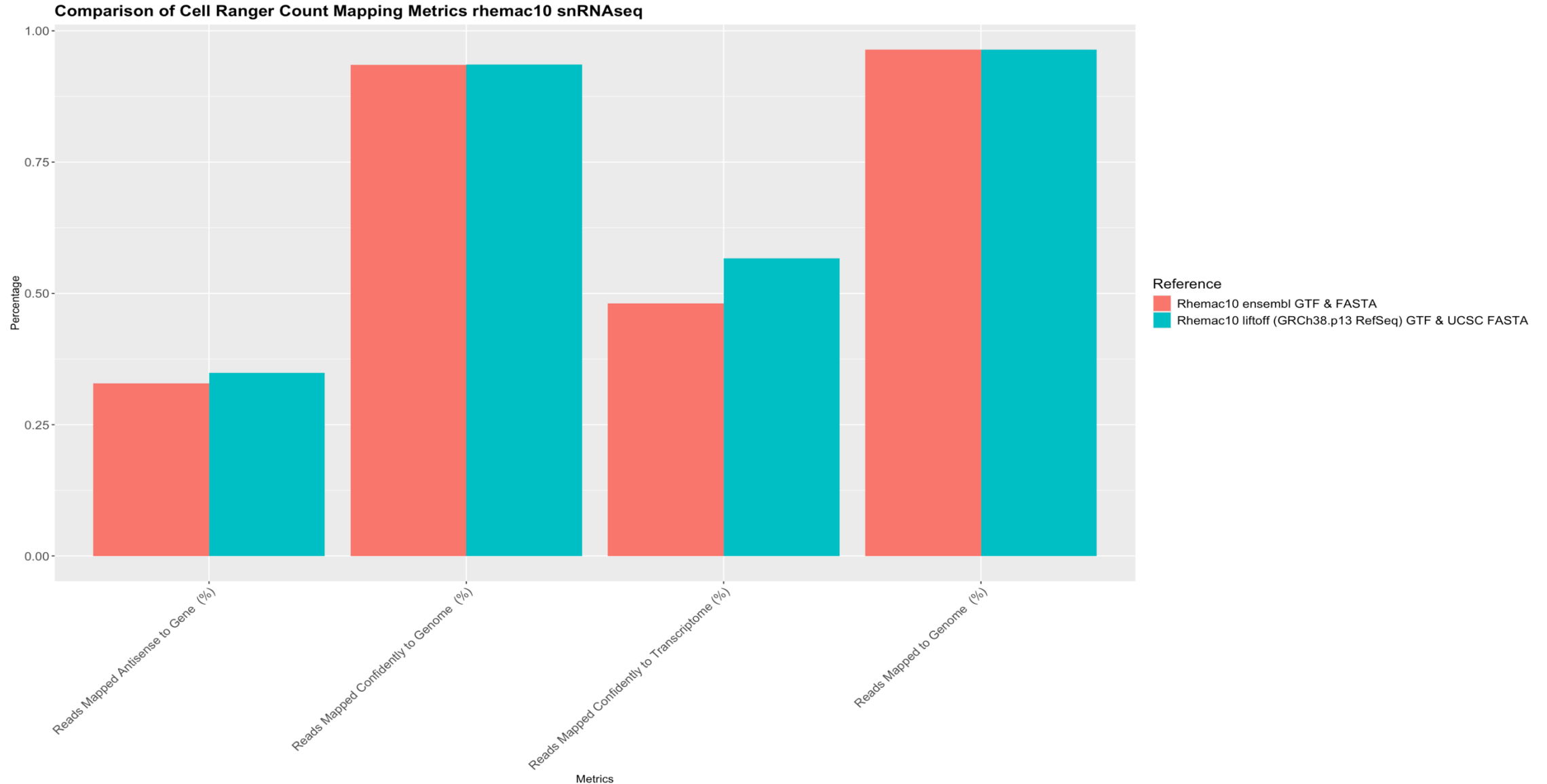
1. Run Cell Ranger

- With rhemac10 liftoff GTF & UCSC FASTA, run Cell Ranger software from 10X to map reads to genome and generate count matrices
 - Run [mkgtf pipeline](#) (filters GTF to remove pseudogenes, mitochondrial genes, etc)
 - Run [mkref pipeline](#) (read in filtered GTF file and rhemac10 FASTA file)
 - Run [cellranger count](#) pipeline (generates a count matrix for each of our 7 libraries). Have 14 fastq files paired-end.
 - Processes reads by trimming TSO from 5' end and polyA from 3' end of R2
 - Align paired end reads to macaque genome using STAR
 - Corrects cell barcodes (unique DNA sequences used to identify each cell type). Checks that cell barcodes found match a "whitelist" of known cell barcodes in scRNAseq/snRNAseq data
 - Filter UMI's (Unique Molecular Identifiers) that have contain N's or have bases with a base quality lower than 10
 - Counts UMI's. Only the confidently mapped reads with valid barcodes and UMI's are used for counting.
 - Calls cells which basically finds GEMs/barcodes containing cells and filters out GEMs without cells using EmptyDrops
- Generates filtered feature barcode matrix that contains only detected cell-associated barcodes
- Once have count matrices from cellranger for 7 samples, read each of them into R as separate R objects

Cell Ranger Mapping Stats Rhemac10 Ensembl vs Liftoff



Cell Ranger Mapping Stats Rhemac10 Ensembl vs Liftoff



2. Run SoupX & DropletQC

- Followed initial QC steps outlined in Phan et al. 2024 (Logan Lab)
- On each sample/R object, run the following QC software on count matrices before creating Seurat objects (both didn't remove any nuclei)
 - [SoupX](#) - identifies & removes ambient mRNA reads
 - [DropletQC](#) - identifies & removes empty droplets
- Neither tool removed nuclei
- Raw data
 - 37,242 genes & 69,801 nuclei
- Make Seurat objects from the 7 count matrices

3. Run miQC

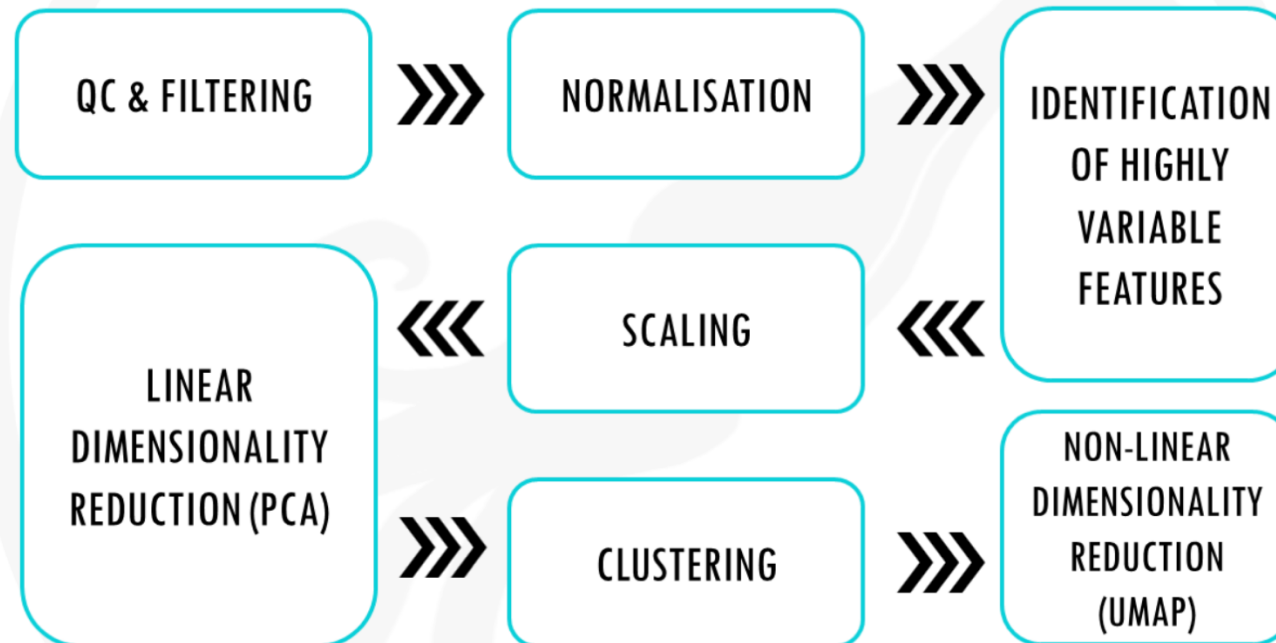
- Calculate percentage of mitochondrial reads (percent.mt) on each Seurat object before running miQC
 - miQC requires 2 metrics per sample to detect damaged nuclei (1) number of unique genes detected per cell (n_Feature_RNA) and (2) percent mitochondrial reads (percent.mt)
- Identify damaged nuclei by running [miQC](#)
 - miQC discards all cells with greater than or equal to 75% probability being compromised. A cell is compromised if it has greater than 10% of cell counts mapping to mtDNA (mitochondrial DNA)

4. Filter out low quality cells in each sample separately

- Subset seurat object meta data to only keep nuclei that are labeled as “good” or “keep”
- Calculate percentage of ribosomal reads (percent.rb) in each sample/filtered seurat object and add to seurat object meta data
- Subset nuclei that have a UMI > 1000, percent.mt < 5%, and percent.rb < 5% to remove low quality cells

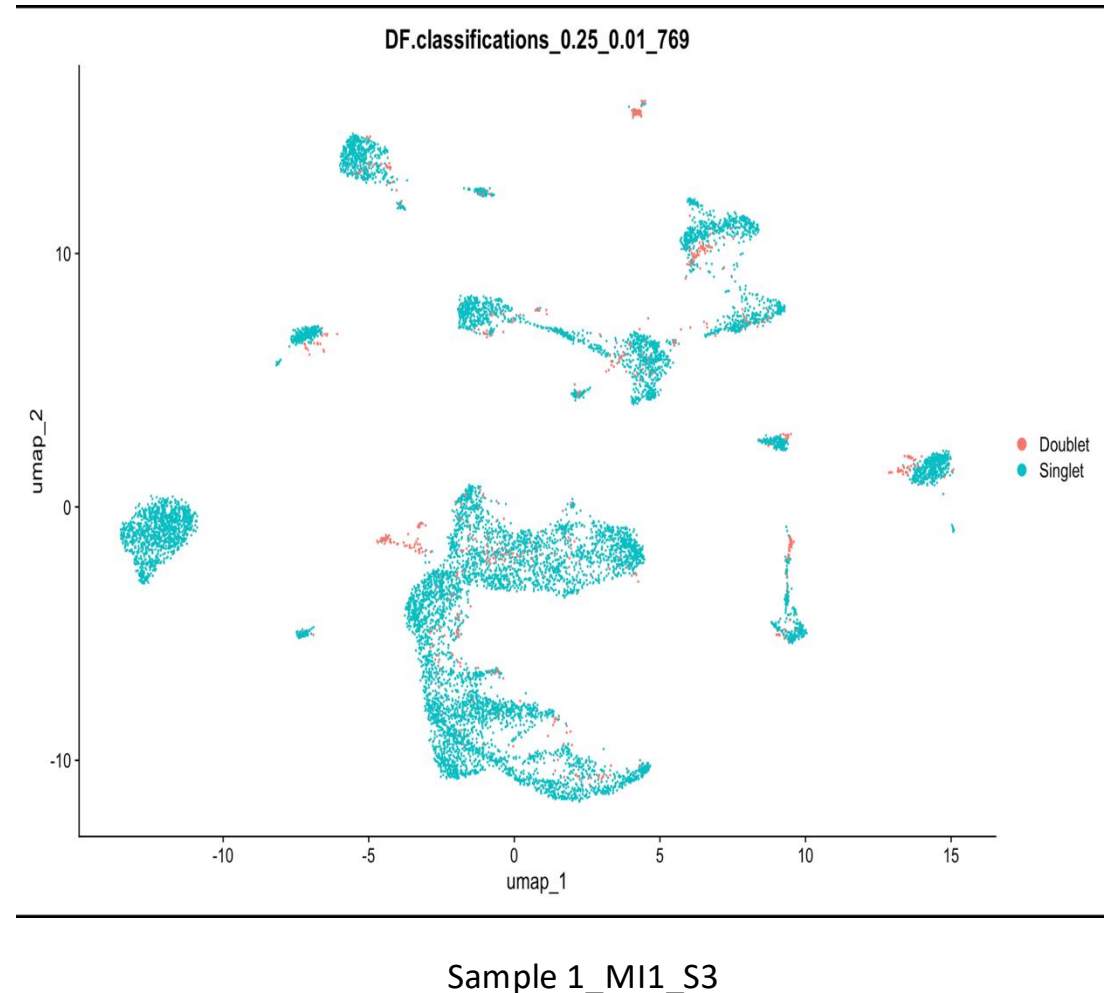
5. Run Seurat on each sample

- Run [Seurat](#) steps before running Doublet Finder (needs clustering information to detect and remove doublets)



6. Run DoubletFinder

- Detected doublets in each sample with [Doublet Finder](#)
- Xi NM & Li JJ Cell Systems 2021 found DF to be the most accurate doublet detection method
- Simulates artificial doublets by averaging gene expression profiles of random pairs of cells.
- Doublet scores of each droplet defined as the cell's proportion of artificial k -nearest neighbors in PC space.
- Final doublet classifications assigned to cells with the n (total number of expected doublets) highest pANN value (cell's proportion of artificial nearest neighbors)
- Removed doublets from all samples

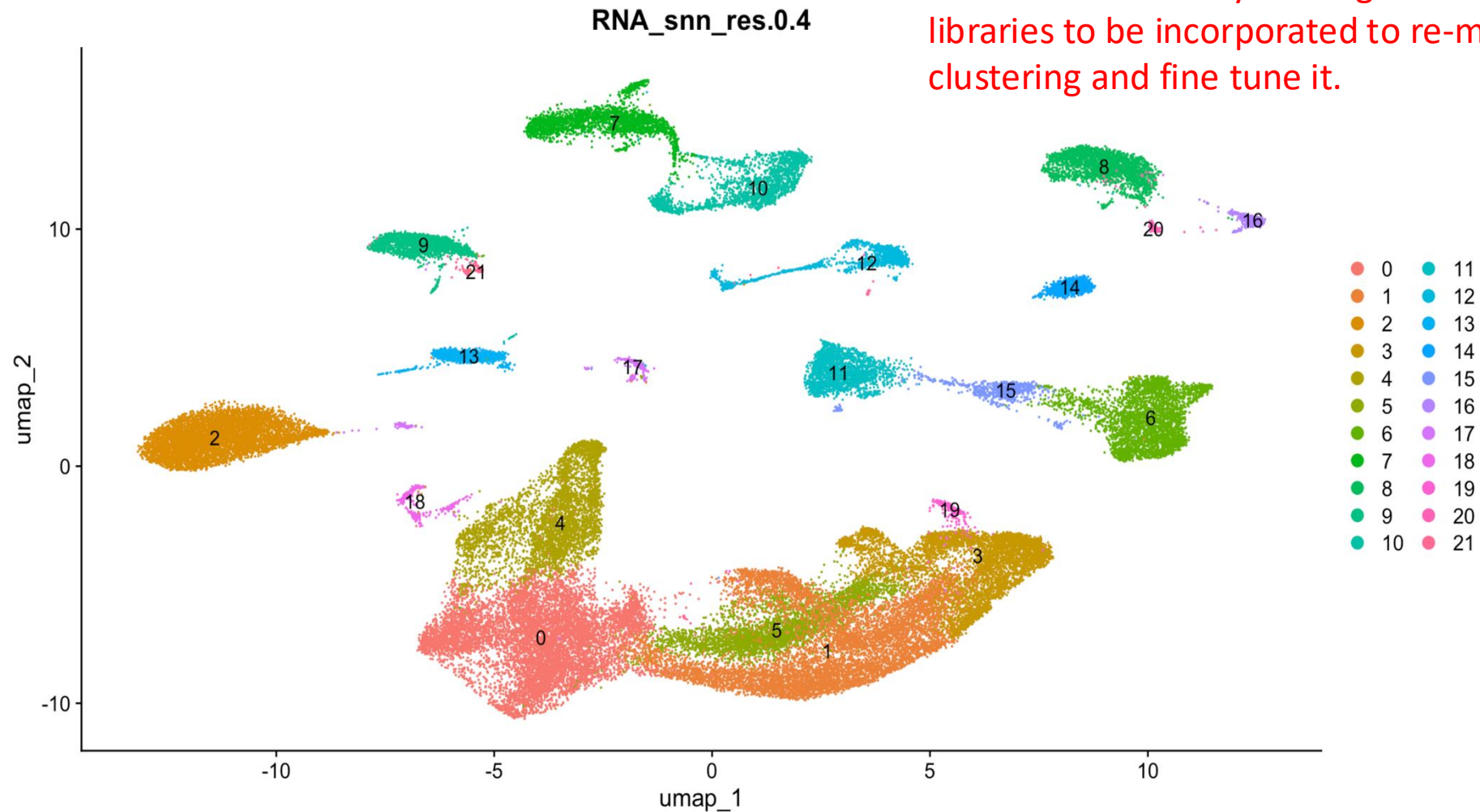


7. Combine samples

1. Create one combined seurat object
 - a) Create list (.csv file) of cell barcodes from each sample seurat object that passed QC (will have 7 csv files)
 - b) Add sample prefix to each cell barcode (example: 1_MI1_S3_AAACCCAAGAGACAAG-1)
 - c) Once you have added sample prefixes, concatenate all 7 csv files into one csv file and name it “good cells” or “cells that passed QC”
 - d) Go back to original count matrices of each sample (cellranger count output). Make a copy of each one. Add sample prefixes to each sample count matrix.
 - e) Merge all 7 count matrices together to make one combined count matrix (cell barcodes labeled by sample)
 - f) Read in count matrix into R and create seurat object with it
2. Use seurat filter function to filter combined seurat object using the csv file make above in c
 - a) 56,020 nuclei total after QC completed
3. Run Seurat pre-processing steps to create one UMAP

Combined sample UMAP

***We know this is not high enough resolution, but we are currently waiting for additional libraries to be incorporated to re-make clustering and fine tune it.



```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
8477 7848 5737 4988 4128 3604 3501 2634 2563 2150 2100 2078 1551 1301 861 810 431 422 286 214 199 137
[1] 56020
```

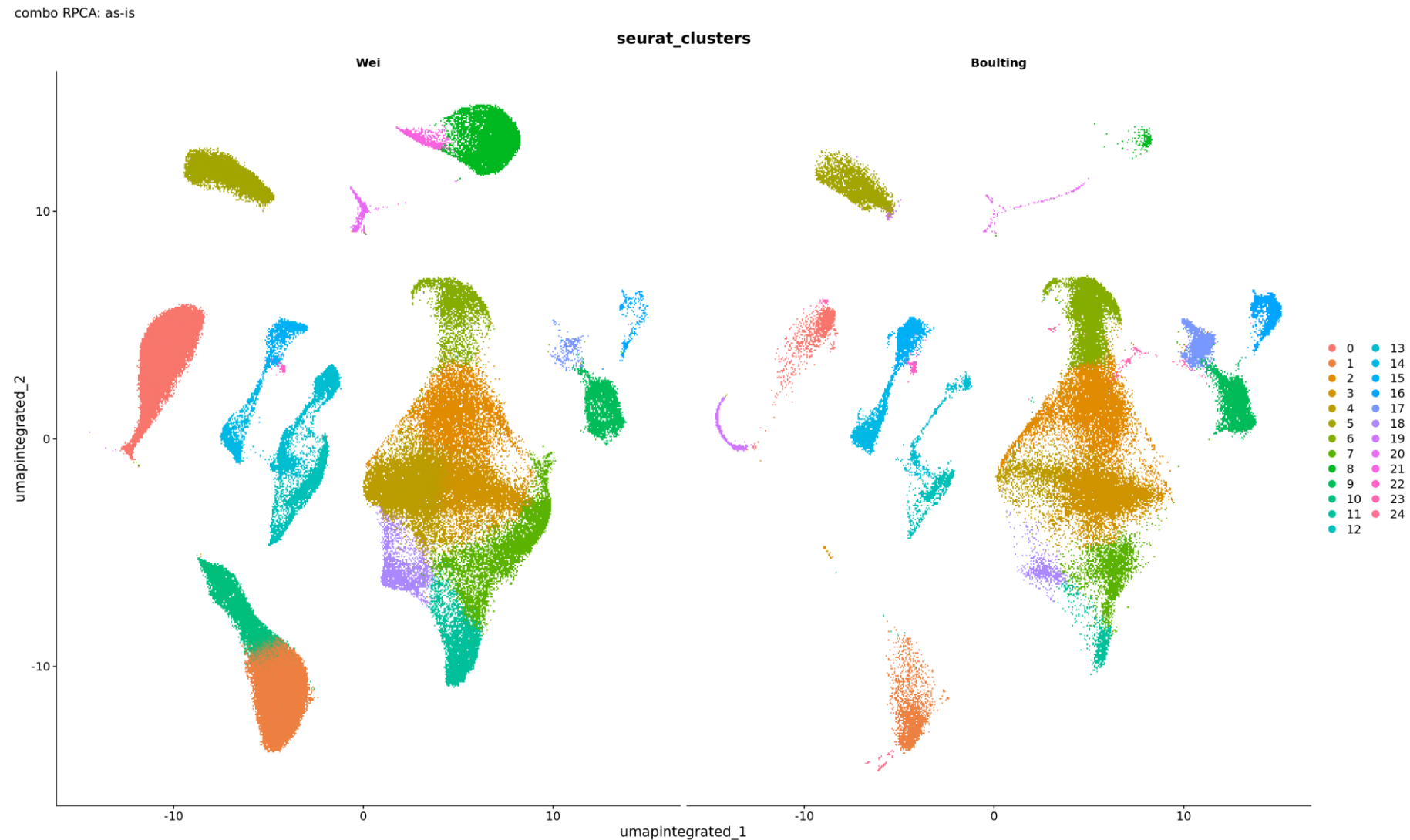
Integration with published data (maybe overkill?)

- Want to validate the quality of our data and assess the cell types we are finding by integrating with the other existing macaque V1 data set.
- Wei et al. 2022: <https://www.nature.com/articles/s41467-022-34590-1>

DOI: <https://doi.org/10.1038/s41467-022-34590-1>

*Note: Wei et al used *macaque fascicularis* (not *mulatta*) so we just used homologous genes

Initial integration with Wei at al. 2022 looks OK, but the grad student left the lab so we need to start over.



Predicting Cell states: “Active” v “Inactive” nuclei

Utilized bulk RNA-seq from rhesus macaques

- Concatenated FPKMs for canonical ADGs
 - Calculated average and min
 - Most promising: **read count cut off = $\lfloor (\text{min FPKM})/10 \rfloor + 1$**
 - **Green** = canonical ADGs
 - **Pink** = novel ADGs with similar expression to canonical ADGs and identified as candidate ADGs in literature
 - Ataman et al., 2016
 - Hvartin et al., 2018
 - **Purple** = novel ADGs with similar expression to canonical ADGs

	AVG FPKM	min FPKM	cut off (avg)	cut off (min)	$\lfloor \text{min FPKM} / 10 \rfloor + 1$
gene-BDNF	2.72	1.89	2	1	1
gene-EGR1	23.54	13.77	23	13	2
gene-FOS	6.24	3.61	6	3	1
gene-LINC00473	15.17	10.69	15	10	2
gene-NPAS4	1.17	0.45	1	0	1
gene-OSTN	12.56	1.02	12	1	1
gene-PER1	13.23	8.51	13	8	1
gene-ZNF331	5.93	5.31	5	5	1
gene-NR4A3	1.54	1.17	1	1	1
gene-HSD17B12	28.19	19.48	28	19	2
gene-PCSK1	54.95	26.52	54	26	3
gene-CYTOR	2.43	0.69	2	0	1
gene-NEFM	51.84	35.84	51	35	4
gene-COL25A1	3.68	2.42	3	2	1
gene-PDE4B	25.57	20.64	25	20	3
gene-ETV5	30.72	24.64	30	24	3
gene-VGF	17.68	7.95	17	7	1

```
total cells: 39702
half cells: 19851
```

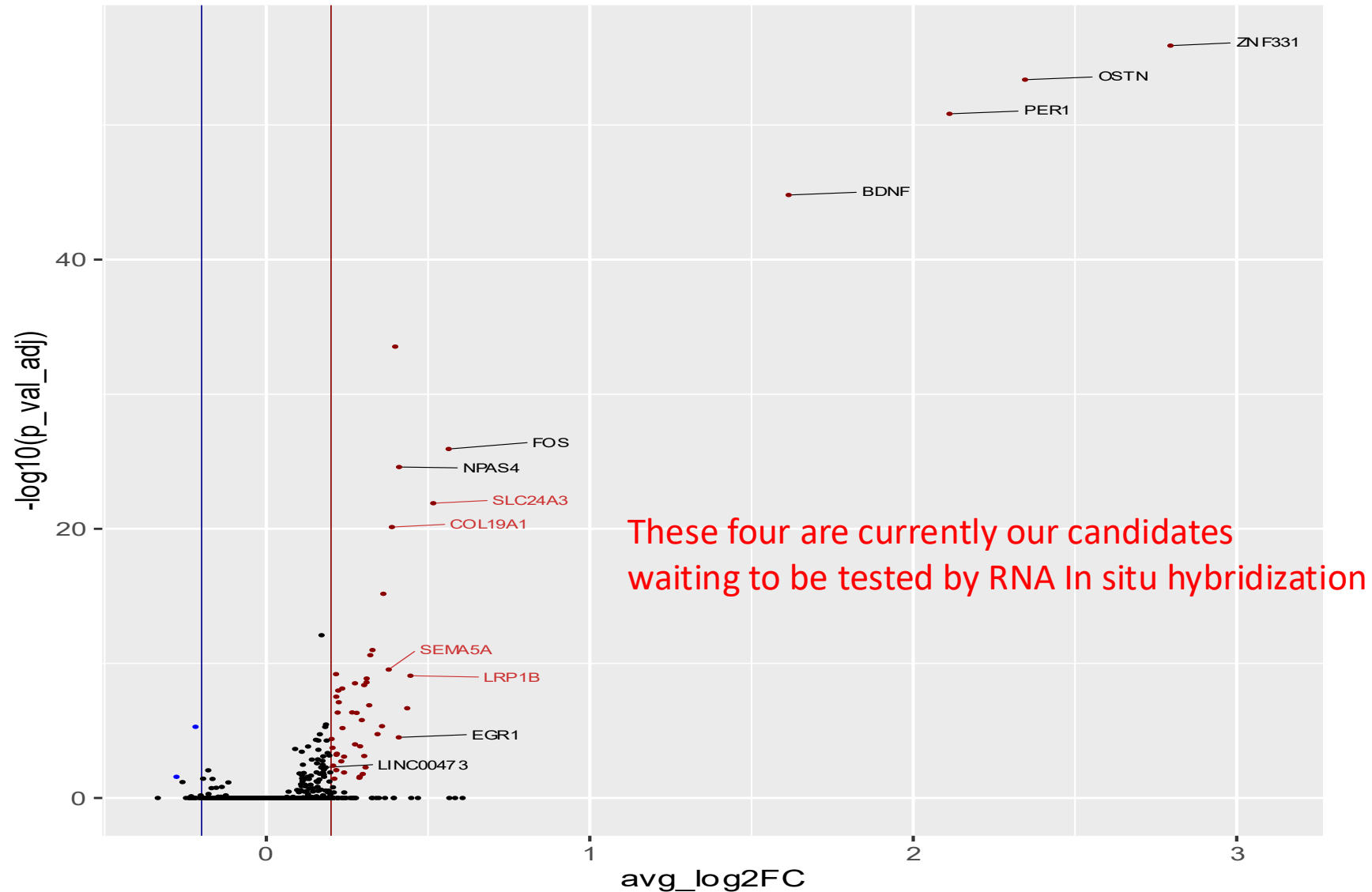
```
Filter parameter for active cell ID: [at least 1 read of any high-confidence gene marker for an active cell]
active cells: 16379
% of cells: 41.3%
```

```
Filter parameter for active cell ID: [min FPKM in [1-9], at least 1 read | min FPKM in [10-19], at least 2 reads | ...]
active cells: 13839
% of cells: 34.9%
```



Pseudobulking,
DGE

Predicting Cell states: “Active” v “Inactive” nuclei



Genome assembly & annotation we will use moving forward

- Assembly: rheMac10 ([UCSC download](#))
- Annotation: [rheMac10-hg38-gencode.v47.basic.gtf.gz](#)

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